# Similarity-metrics for fast retrieval of manoeuvrer data reduced to one dimension using Space Filling Curves - Supplementary Materials

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| Type | Manoeuvre description | L2 | Pearson | DTW |
|---|---|---|---|---|
| 1 | Right lane change, speed up | 0.4975 | 0.4975 | 0.3923 |
| 2 | Right lane change, slow down | 0.7809 | 0.7809 | 0.7809 |
| 3 | Left lane change, speed up | 0.9107 | 0.9107 | 0.8603 |

TABLE I

MCC FOR EACH TYPE OF MANOEUVRE IN REAL DATA DEPENDING ON THE SIMILARITY METRIC USED.

| | L2 | Pearson | DTW |
|---|---|---|---|
| Accuracy | 0.8915 | 0.8915 | 0.8760 |
| Precision | 0.8372 | 0.8372 | 0.8140 |
| Recall | 0.8372 | 0.8372 | 0.8140 |
| MCC | 0.7558 | 0.7558 | 0.7209 |

TABLE II

CLASSIFICATION OF ALL MANOEUVRES FOR EACH TYPE OF MANOEUVRE IN REAL DATA DEPENDING ON THE SIMILARITY METRIC USED.

## I. DOWNSHIFTING AND NORMALISATION

First, there is a pre-processing step in which the data is filtered to reduce noise. This is something we learned was important when working with real data. Only after that is the data encoded using SFCs. After encoding, we include some post-processing steps in order to better differentiate between the types of manoeuvres, as showcased in Fig. 1: normalisation and downshifting.

## II. REAL DATA

Figures 2 to 10 and 8 to 16 are introduced here to illustrate the impact that each input signal (steering wheel angle, and speed) have on the overall shape of the original and shifted characteristic stripe patterns (CSPs). Despite the representation in these figures, it is important to note that the original signals can't take negative values when encoding them with the *ZCurve* Python library. The values in these figures are therefore have been mapped around 0 purely for visualisation purposes, given that the original signals and the original and shifted CSPs are not in the same range.

For the real data, speeding up(down) is classified as being manoeuvrers where the average speed taken over the last ten time steps is higher(lower) compared to the average taken over the first ten time steps. Using this as a classification may cause significant speed changes during the main part of the maneuvers to be neglected, resulting in misleading annotations.

Downshifting does not have a large impact on the shape of the csps in figures 2 - 4, because there are no large bit shifts in the data we used.

The reason for using Matthew Correlation Coefficient (MCC) to study the efficacy of our method is class imbalance: each type of manoeuvre has few samples compared to the whole dataset. This is particularly true for synthetic data, which contains 10 different types of manoeuvres, as seen in Table III.

| ID | Manoeuvre description | ED | Pearson | DTW |
|---|---|---|---|---|
| 1 | Right turn, speed up | -0.0534 | 0.0000 | 0.0000 |
| 2 | Left turn, speed up | 1.0000 | 1.0000 | 0.8542 |
| 3 | Right turn, slow down | 0.8542 | 0.8542 | 0.6667 |
| 4 | Left turn, slow down | 1.0000 | 1.0000 | 0.8819 |
| 5 | No turns, speed up | 0.4583 | 0.6186 | 0.8819 |
| 6 | No turns, slow down | 0.7935 | 0.7935 | 1.0000 |
| 7 | Right lane change, speed up | 0.8542 | 0.8542 | 0.8542 |
| 8 | Left lane change, speed up | 1.0000 | 1.0000 | 0.7222 |
| 9 | Right lane change, slow down | 0.8542 | 0.8542 | 1.0000 |
| 10 | Left lane change, slow down | 1.0000 | 1.0000 | 0.7222 |

TABLE III

MCC FOR EACH TYPE OF MANOEUVRE IN SYNTHETIC DATA DEPENDING ON THE SIMILARITY METRIC USED.

As you can see in Table I (as well as in Tables II, III, and IV), many of the values for the MCC, accuracy, precision, and recall have the same values for the three distance metrics. This is due to chance and the low number of samples used in the evaluation.

## III. SYNTHETIC DATA

The sampling rate for the synthetic data is approximately four times higher compared to the real data. This may affect the evaluation scores.

In Figure 9 we see the effect of downshifting. If the CSP is not downshifted, the speed increase overpowers the contribution from the angle.

## ACKNOWLEDGMENT

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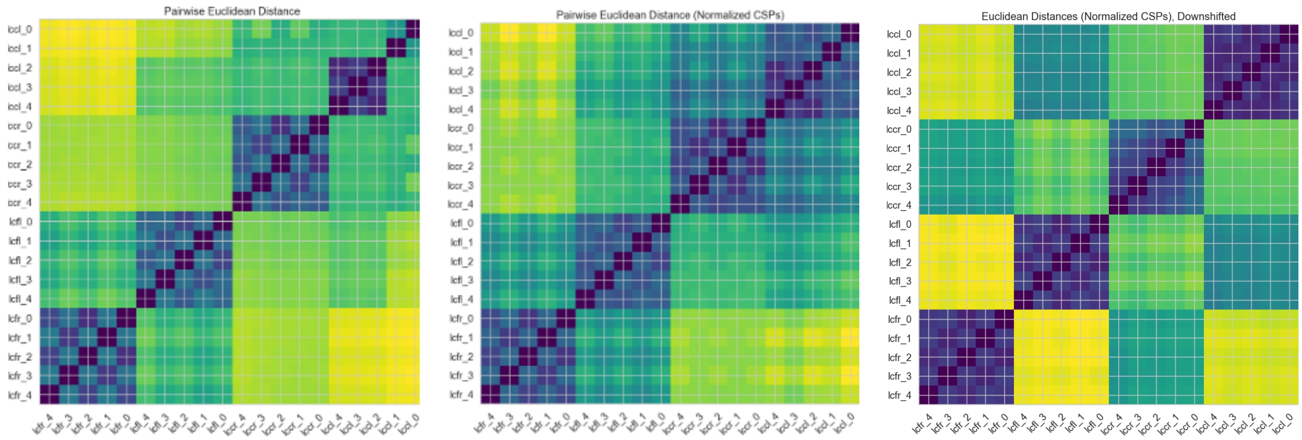


(a) No CSP post-processing. (b) Normalisation. (c) Normalisation and downshifting.

Fig. 1. Relative similarity based on euclidean distances between data samples (in a scale from dark blue, indicating 100% similarity, to yellow, indicating the opposite), computed with different CSP post-processing steps. The samples are named as follows: "lc" stands for lane change, so they are all this type of manoeuvre; then, "c" indicates a manoeuvre that starts early in the studied time window, while "f" indicates that it starts later; finally, "l" stands for left lane changes, and "r" indicates one towards the right. This figure shows how the post-processing steps help differentiate between the types of manoeuvres.

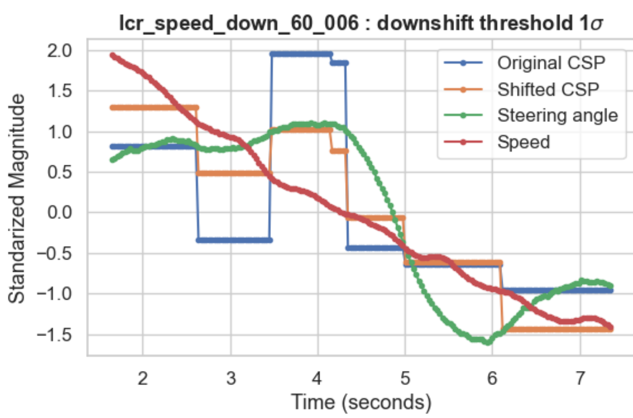


Fig. 2. CSP (original and shifted) for real data manoeuvre type 1.

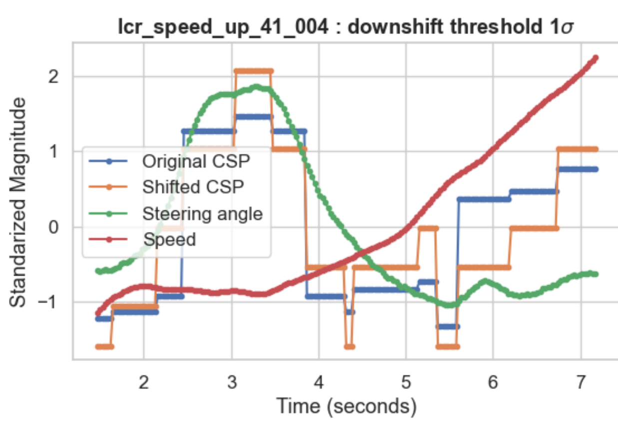


Fig. 3. CSP (original and shifted) for real data manoeuvre type 2.

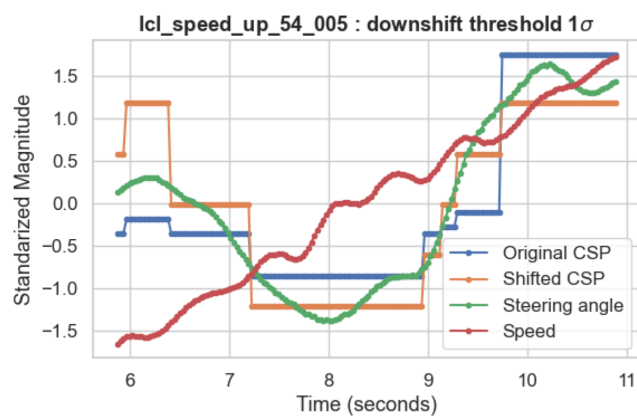


Fig. 4. CSP (original and shifted) for real data manoeuvre type 3.

| | ED | Pearson | DTW |
|---|---|---|---|
| Accuracy | 0.9600 | 0.9650 | 0.9600 |
| Precision | 0.8000 | 0.8250 | 0.8000 |
| Recall | 0.8000 | 0.8250 | 0.8000 |
| MCC | 0.7778 | 0.8056 | 0.7778 |

TABLE IV
CLASSIFICATION OF ALL MANOEUVRES FOR EACH TYPE OF MANOEUVRE IN SYNTHETIC DATA DEPENDING ON THE SIMILARITY METRIC USED.

computations and data handling were enabled by resources provided by the National Academic Infrastructure for Supercomputing in Sweden (NAISS), partially funded by the Swedish Research Council through grant agreement no. 2022-06725.

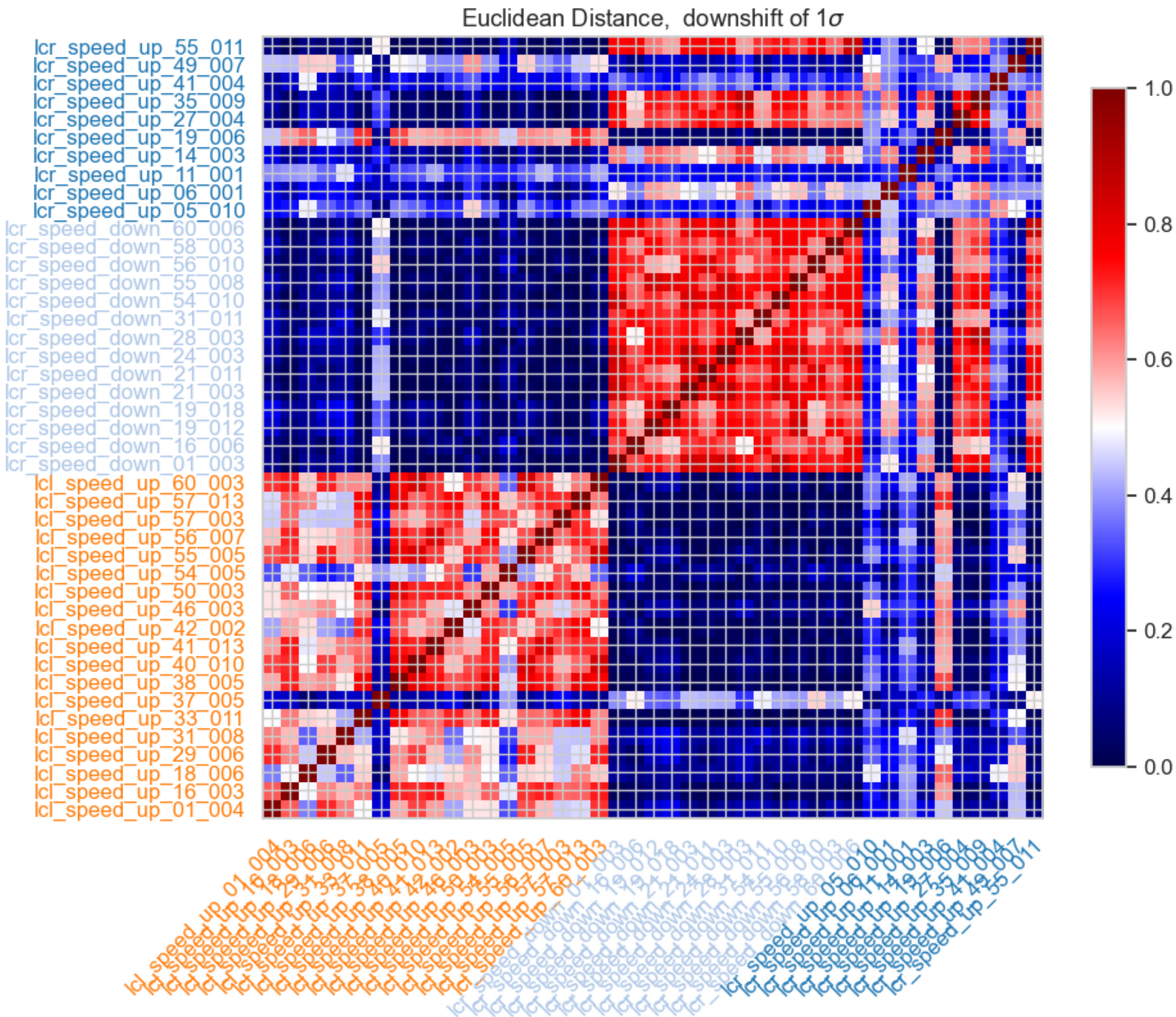


Fig. 5. Heat map for real data classified based on Euclidean distance

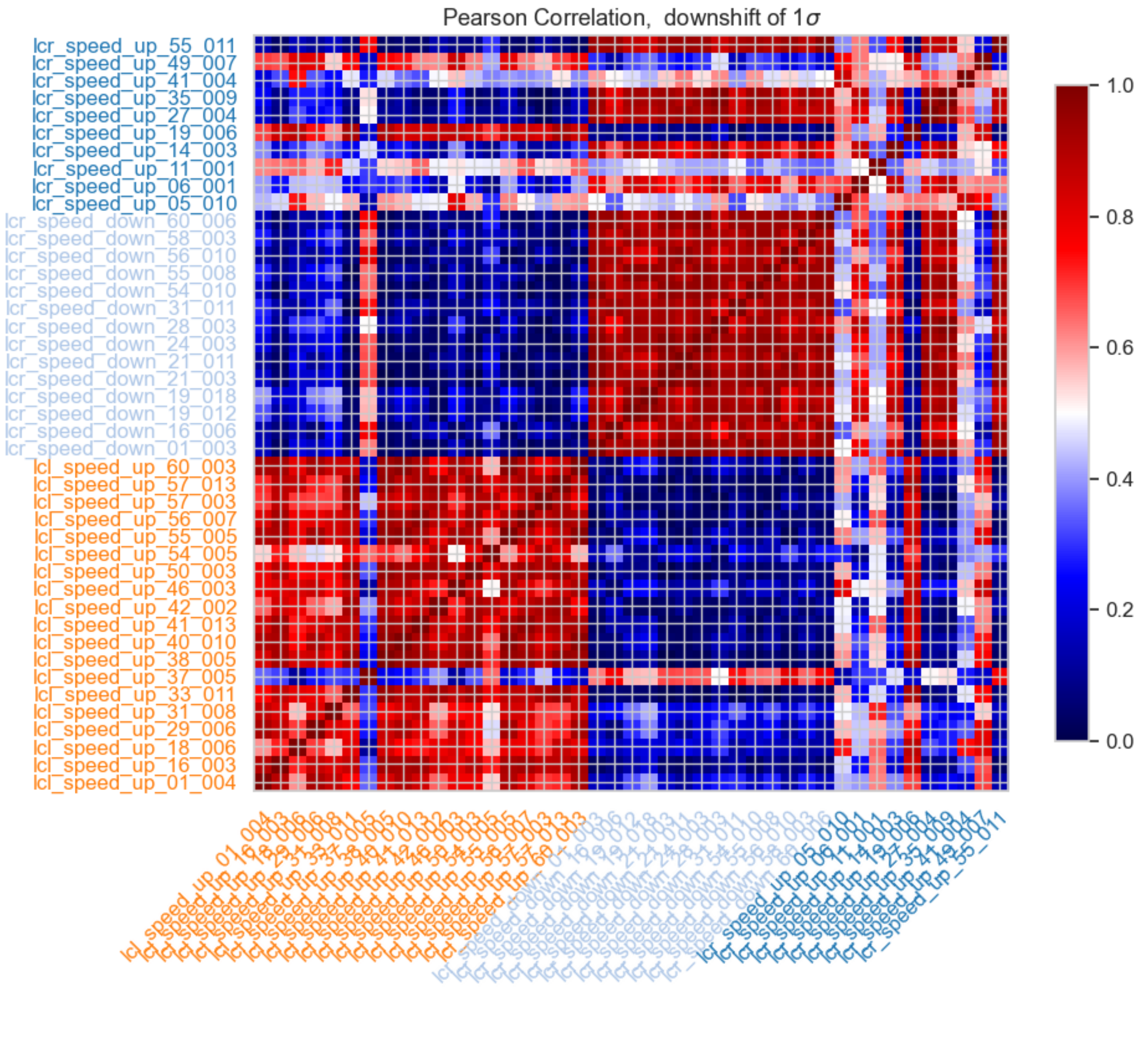


Fig. 6. Heat map for real data classified based on Pearson coefficient-based distance

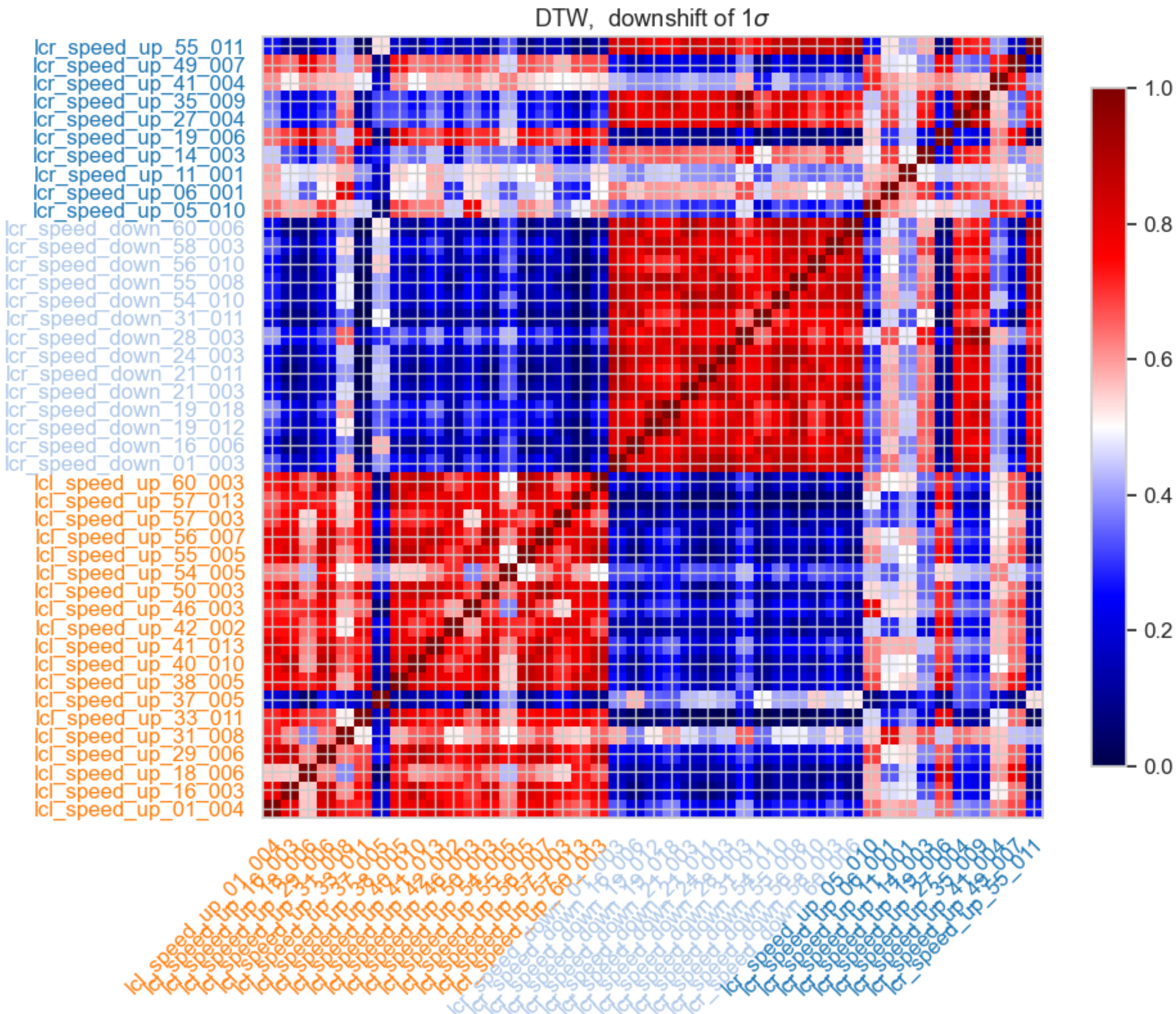


Fig. 7. Heat map for real data classified based on DTW distance

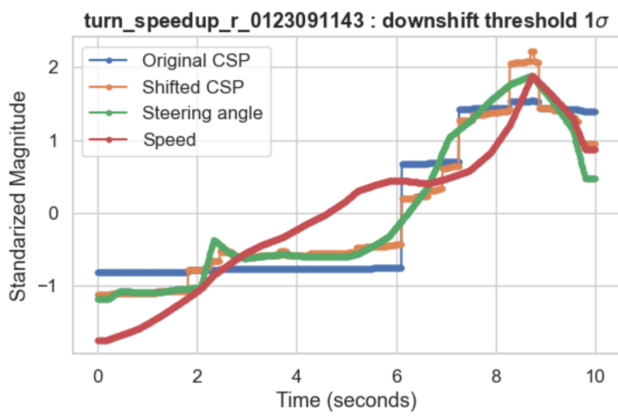


Fig. 8. CSP (original and shifted) for synthetic data manoeuvre type 1.

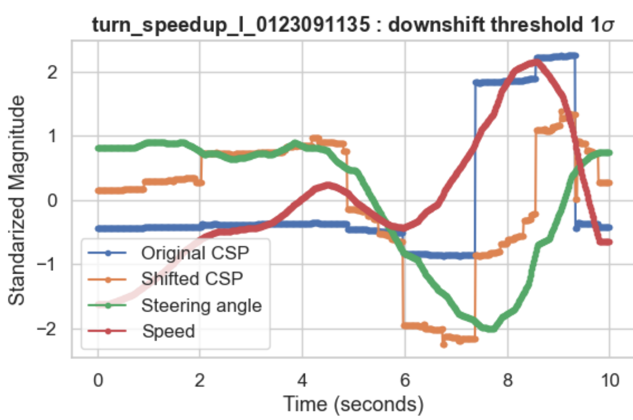


Fig. 9. CSP (original and shifted) for synthetic data manoeuvre type 2.

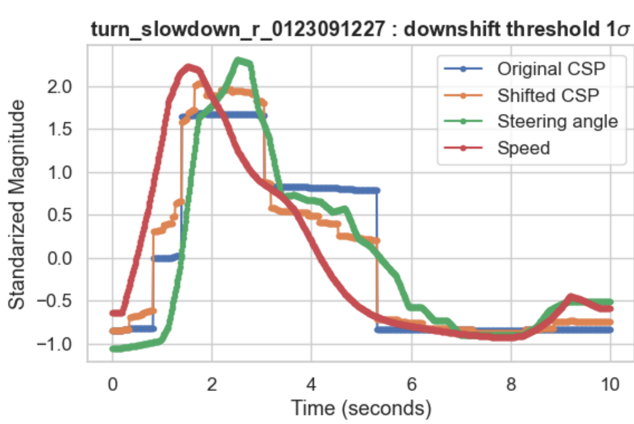


Fig. 10. CSP (original and shifted) for synthetic data manoeuvre type 3.

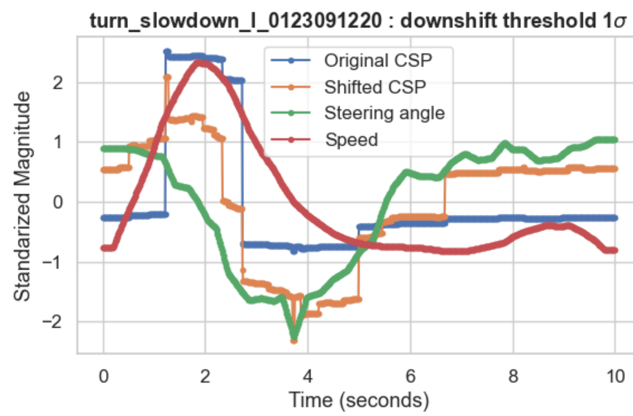


Fig. 11. CSP (original and shifted) for synthetic data manoeuvre type 4.

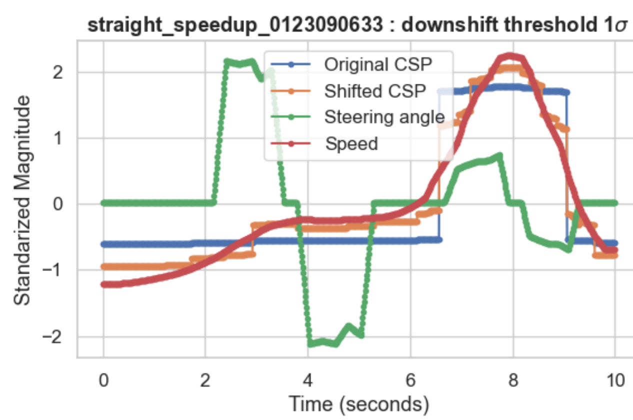


Fig. 12. CSP (original and shifted) for synthetic data manoeuvre type 5.

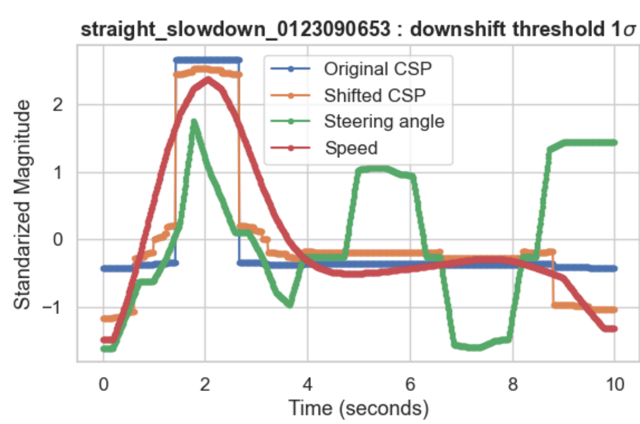


Fig. 13. CSP (original and shifted) for synthetic data manoeuvre type 6.

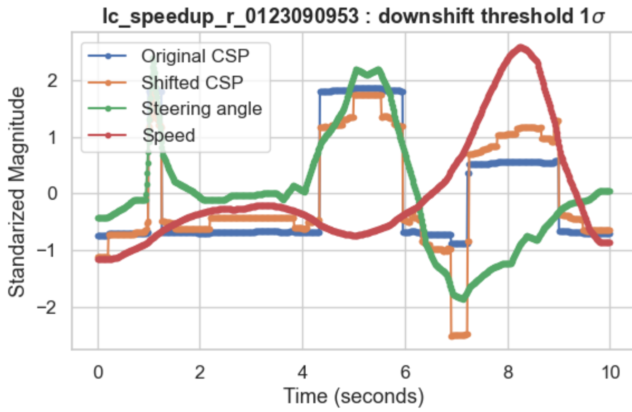


Fig. 14. CSP (original and shifted) for synthetic data manoeuvre type 7.

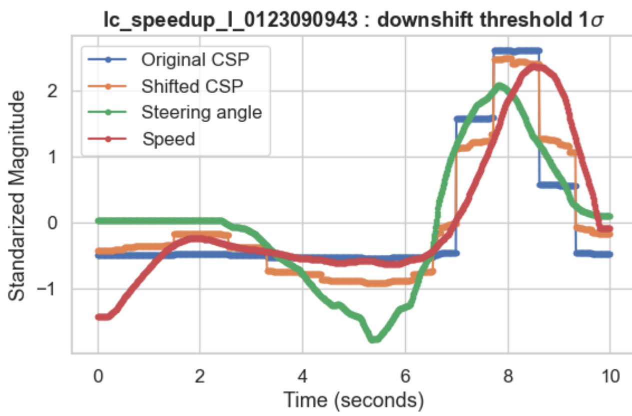


Fig. 15. CSP (original and shifted) for synthetic data manoeuvre type 8.

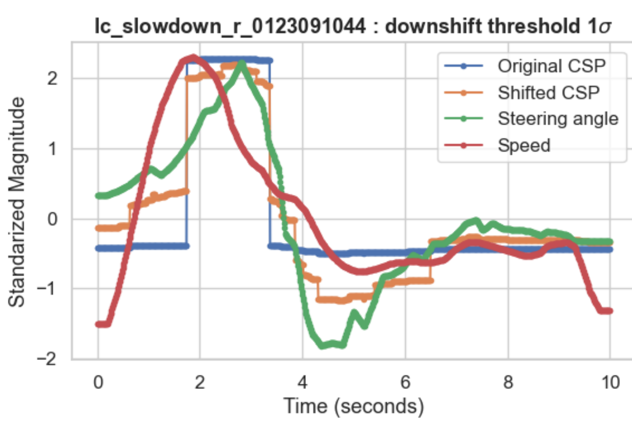


Fig. 16. CSP (original and shifted) for synthetic data manoeuvre type 9.

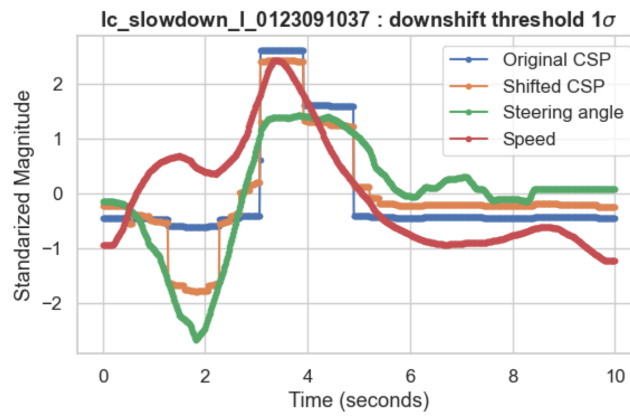


Fig. 17. CSP (original and shifted) for synthetic data manoeuvre type 10.

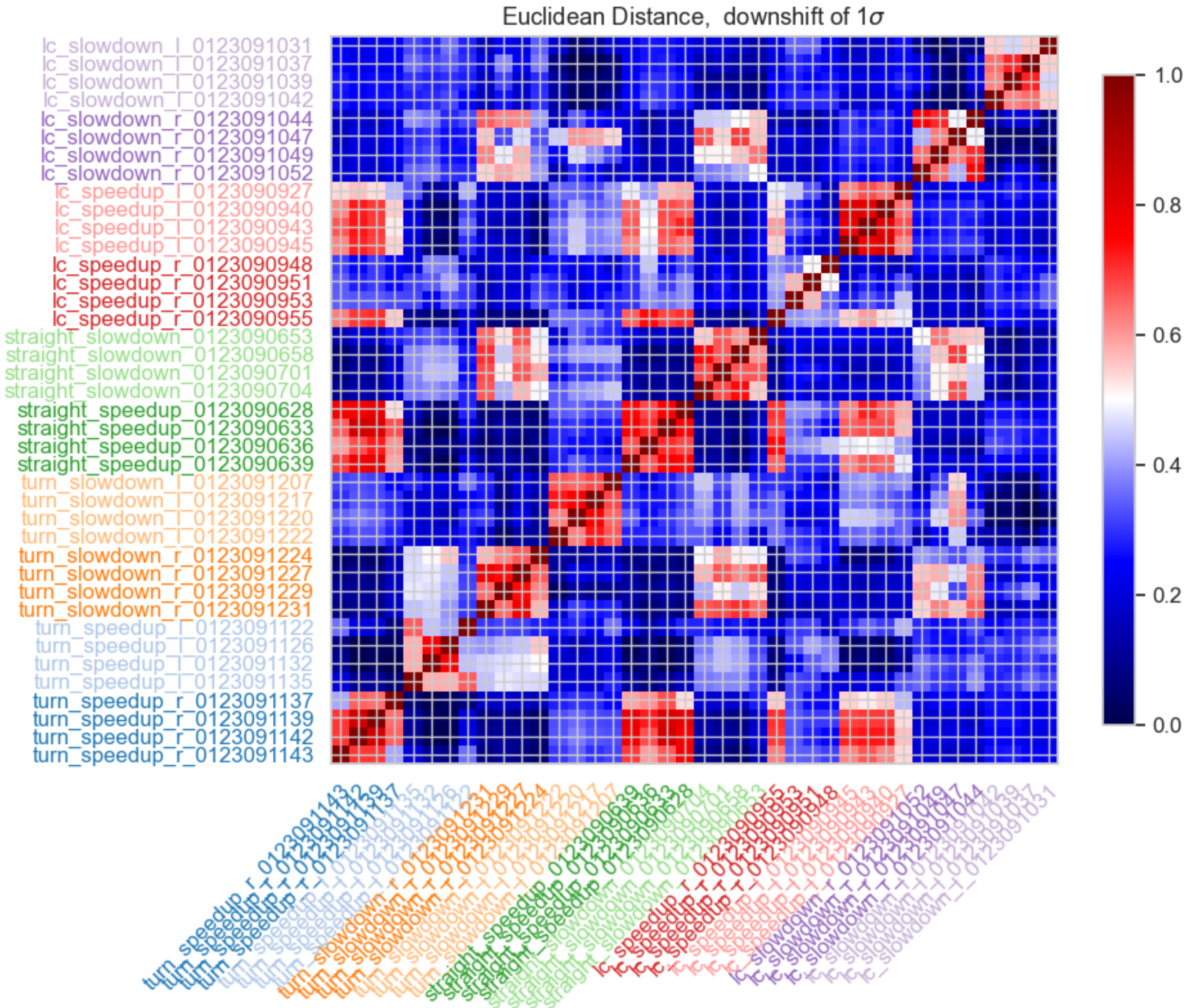


Fig. 18. Heat map for synthetic data classified based on Euclidean distance

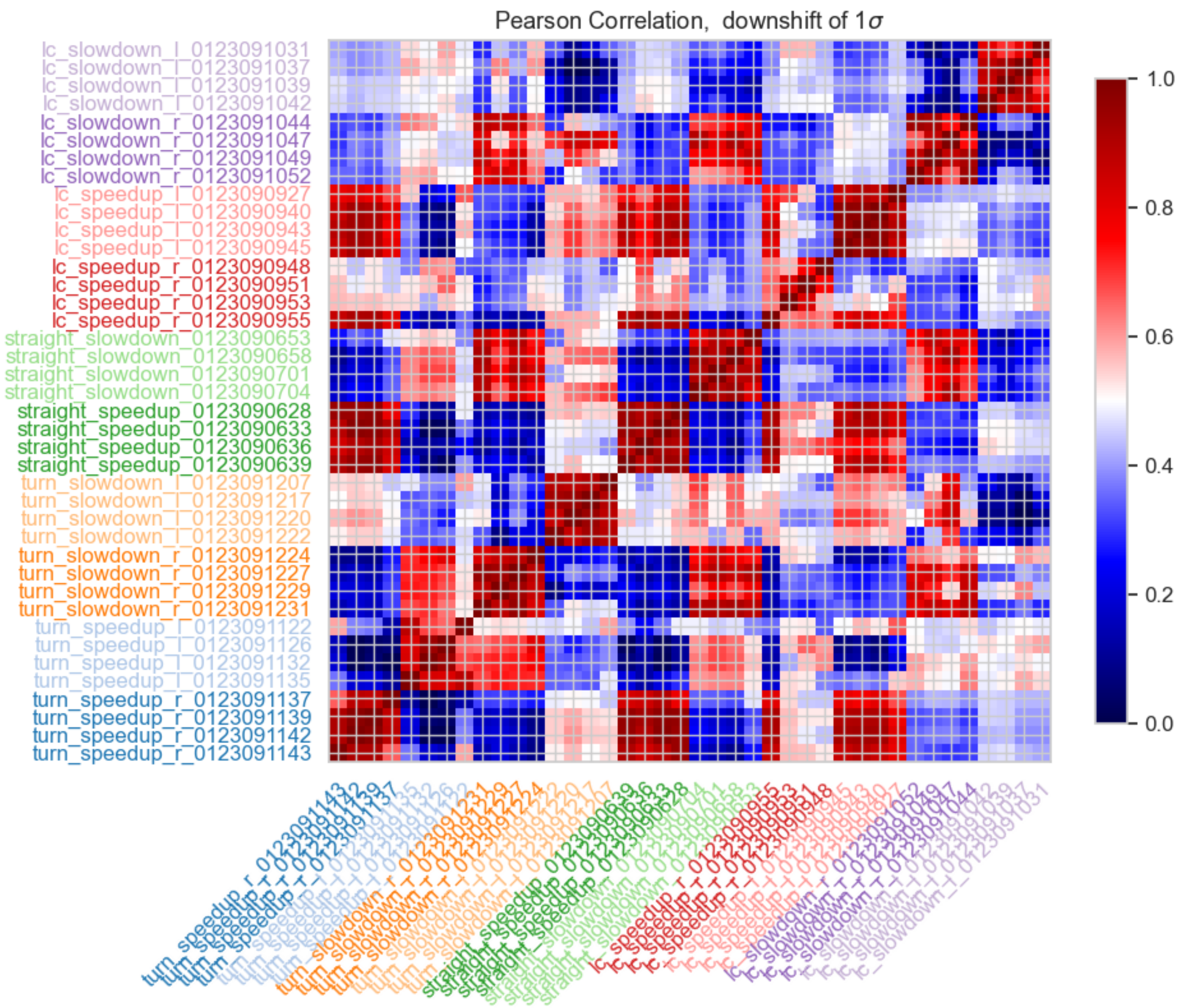


Fig. 19. Heat map for synthetic data classified based on Pearson coefficient-based distance

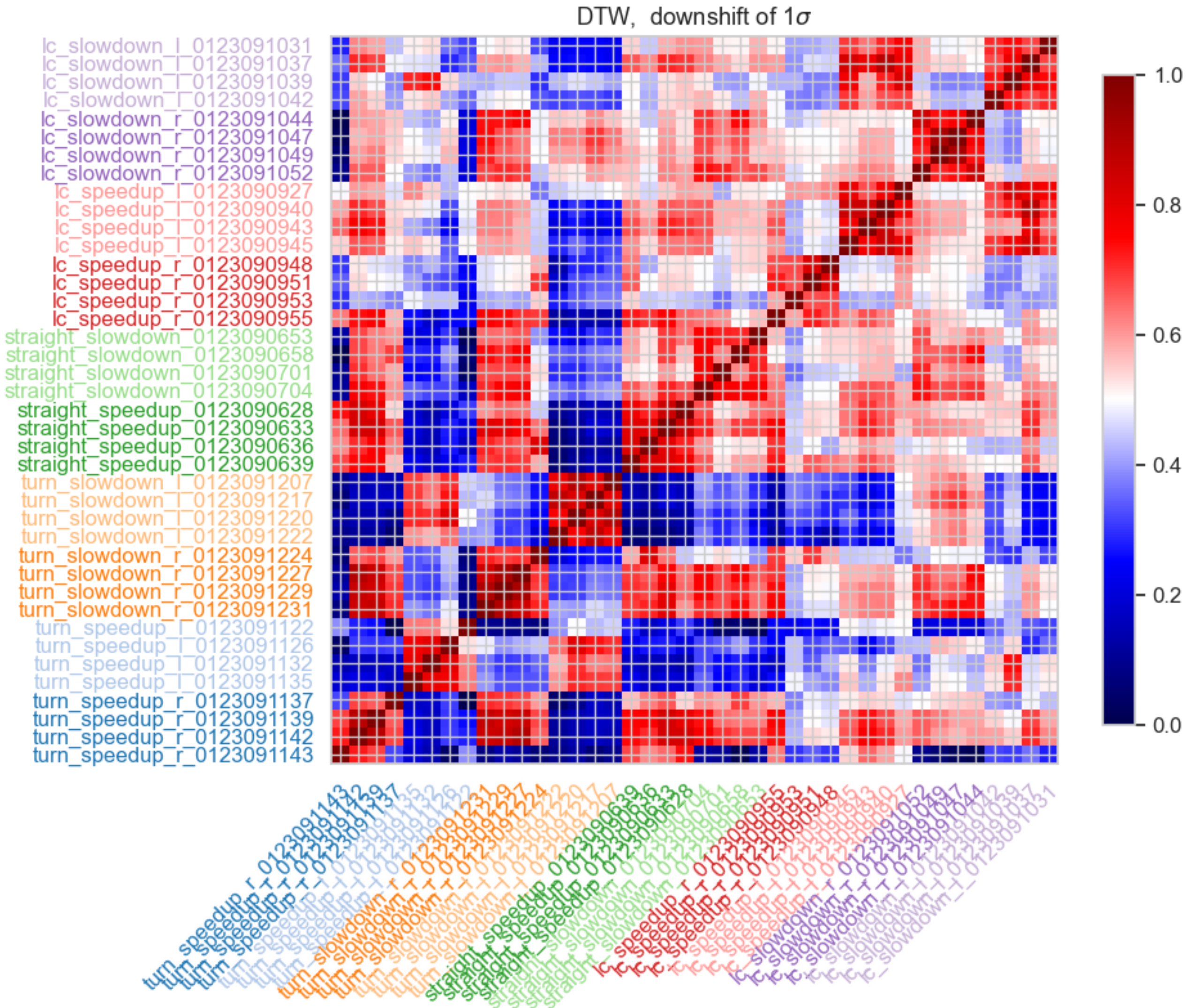


Fig. 20. Heat map for synthetic data classified based on DTW distance